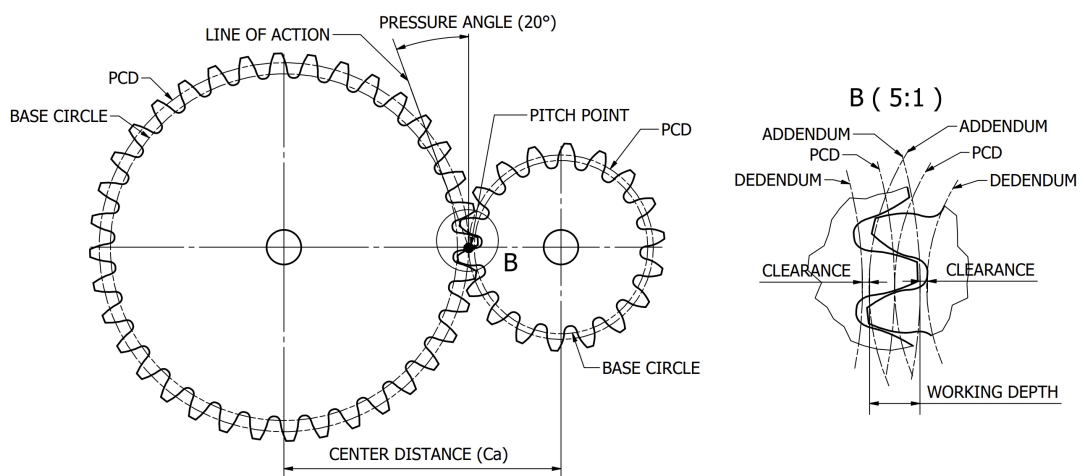
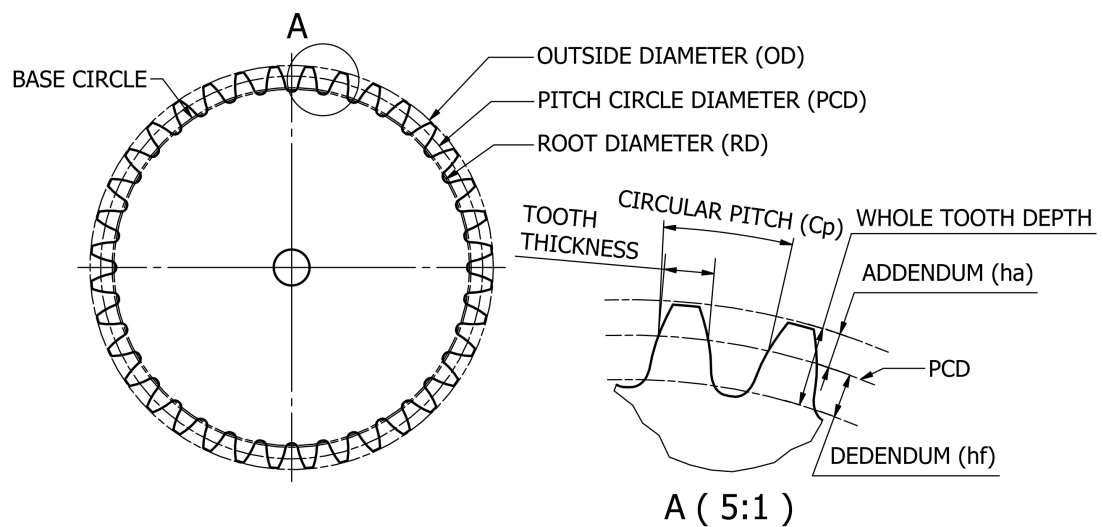


MST THEORY

Eccentric & Planetary Gears

Gear Terminology



$$PCD = mT \Rightarrow m = \frac{PCD}{T}$$

$$h_a = m$$

$$h_f = 1.25m$$

$$OD = PCD + 2h_a$$

$$RD = PCD - 2h_f$$

$$\text{whole tooth depth} = h_a + h_f$$

$$C_p = \pi m = \pi \frac{PCD}{T}$$

$$\text{tooth thickness} = \frac{1}{2} C_p = \frac{1}{2} \pi m$$

$$C_a = \frac{1}{2} (PCD_{\text{driver}} + PCD_{\text{driven}})$$

$$\text{clearance} = h_f - h_a = 0.25m$$

$$\text{working depth} = 2h_a = 2m$$

Advantages of Planetary Gears

Input and Output shafts are coaxial

Higher torque capacity (load is shared between 3 gears)

Compact

Inertially balanced

Disadvantages of Planetary Gears

Complex mechanism

Higher precision manufacturing

Higher cost

Planetary Gear Uses

Fixed Ring, Sun to Carrier

Torque increase, used in robotic manipulators

Fixed Sun, Carrier to Ring

Speed increase and torque reduction, used in car auto transmission gearbox

Fixed Sun, Ring to Carrier

Speed decrease and torque increase, used in electric screwdrivers and bulldozer power trains

Fixed Carrier, Sun to Ring

Speed decrease and torque increase, used in motorcycle transmission for reversing

Keys

Key Types

transmits torque, resists compressive and shear stresses

Sunk Keys	General purpose and most common	
Saddle Keys	Light unidirectional Loads	
Tangent Keys	Slow bidirectional high torque	
Round Keys	low torques and speeds	

Splines

Transmit torque by having teeth on the spline act as keys

Straight Sided Splines (ISO 14)!!!!!!!!!!!!!!!!!!!!

straight tooth profile

number of teeth X minor diameter X outer diameter

Involute Metric Splines (ISO 4156)!!!!!!!!!!!!!!!!!!!!

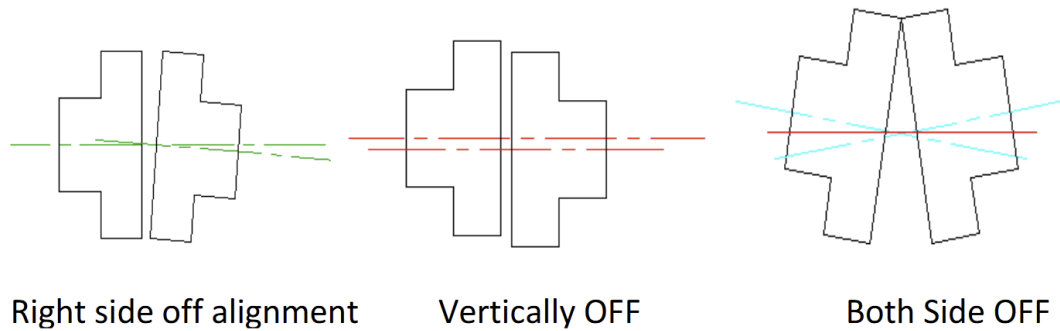
involute curve tooth profile, similar to gear teeth (30 DEG PRESSURE ANGLE)

two types: flat root and fillet roots

Couplings

facilitate misalignment of two shafts, to ensure power and torque transmission. Also allows for maintenance and vibration reduction

three types of alignment that couplings facilitate:



Shaft Pins

fixed shaft supported components onto shafts; prevents axial movement



Retaining Rings/Circlips

Prevent shaft components from falling off the shaft/(axially positions component), either via grooves in the shaft (external circlips) or hubs (internal circlips)

External circlip



Internal circlip



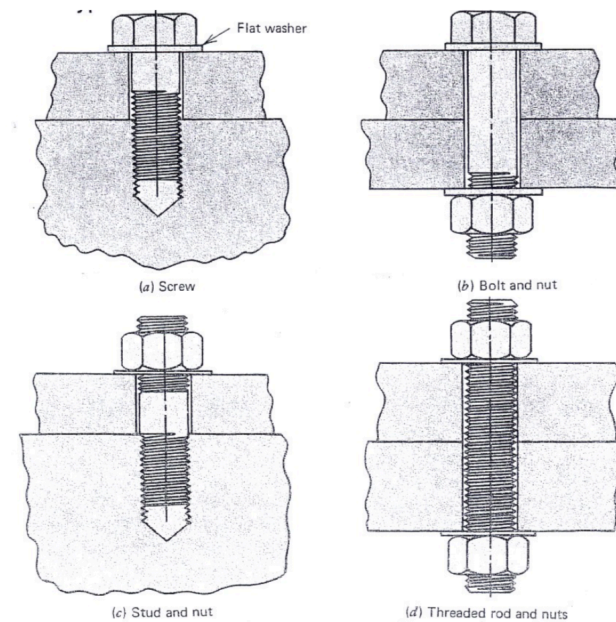
E-clip



Grip ring

Threaded Fasteners

Fastener assembly types



Multiple thread:

good:

larger minor diameter >> better at shear

larger linear movement per rotation

bad:

lesser holding power

Tamper Resistant Heads

prevenet vandalism!!!!!!!!!!!!!!!!!!!!!! used in something idk

Thread Forms

Unified (ANSI/ASME)/Metric Thread(ISO) (Triangular)

single threaded fasteners, threaded pipe fittings and general purpose applications

Unified: Major Diameter X Threads per inch

Metric: Nominal diameter x pitch

Square Thread

High load capacity & accuracy

used in power transmission & machinery

can be used for self locking due to higher friction of threads and prevention of backdriving

ACME thread (trapezium with 29 deg incline)

high efficiency & accuracy for linear motion applications

high strength and durability

used in power screws and jacks

Buttress Thread (sawtooth)

high strength and efficient load transfer in one direction

easy assembly/disassembly in the other direction

Screws for Plastics

self tapping → narrower thread profile (sharper threads)

thread forming/thread cutting

thread forming: pilot hole, symmetric threads, deforms material, lower stiffness

thread cutting: pilot hole, asymmetric threads, cuts material away to form threads, higher stiffness

Bolt Tightening Methods

tighten to torque (common), tighten to angle (difficult to define), tighten to yield point

Bolt tightening Orders

ZigZag tightening technique

Spiral tightening technique

do one pass in 50%, 75% then 100%